

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458074.8198 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.013
Transit depth (Rp/Rs)^2	2.07 +/- 0.37 [%]
Orbital Inclination (inc)	84.72 +/- 0.89
Airmass coefficient 1 (a1)	1.048 +/- 0.01
Airmass coefficient 2 (a2)	-0.0297 +/- 0.0053
Scatter in the residuals of the lightcurve fit is	1.0 %
Best Comparison Star	#1 - [207, 198]
Optimal Aperture	3.77
Optimal Annulus	7.85
Transit Duration (day)	0.0751 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.013 vs 0.1404 (deviation 0.2σ)
Duration fit vs published	0.0751 d vs 0.0754 d (deviation 0.03σ)
Mid-time O–C	3.7 min
Depth SNR	8.1
Residual scatter	1.0 %

Light curve

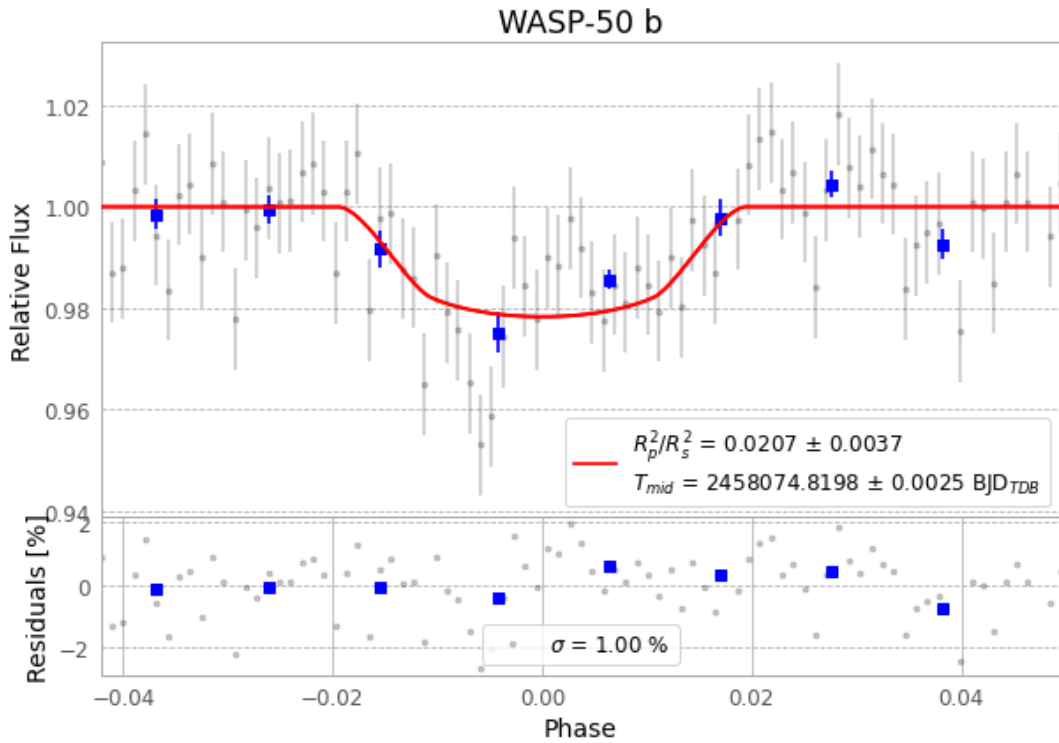


Figure 1 — FinalLightCurve_WASP-50 b_2017-11-16.png (SHA-256 09f0d138b72ef72f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 261], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b6c27cfaf3be6f03...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

87 FITS frames (57 MB), WASP-50171117053312.FITS → WASP-50171117095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-171117.log (b4ff73c5f155...), FinalLightCurve_WASP-50 b_2017-11-16.png (09f0d138b72e...), FinalLightCurve_WASP-50 b_2017-11-16.csv (830b98318b87...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 23:43Z.